Date : 20/10/26
Day : Tuesday

Assessment tool-1
CO_2
Analytical problem solving

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Perceptron model pass or Fail

Given

* Study hours (x_1) = 2
* Attendance (x_2) = 1
* weight of study hours (w_1) = 0.6
* weight of attendance (w_2) = 0.4
* Bias (b) = -0.5

Step 1: Calculate the net input

The net input of a perceptron is calculated by multiplying each input by its corresponding weight and adding the bias.

Net Input = $x_1w_1 + x_2w_2 + b$

Substituting the given

Net Input = $(2 \times 0.6) + (1 \times 0.4)$

= $1.2 + 0.4 - 0.5$

Net Input = 1.1

Apply the step activation function

The step activation function converts the net input into binary output of either 0 or 1

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0 & z < 0 \end{cases}$$

$f(1.1) = 1$

* output 1 represents pass.
* output 0 represents fail.

2. Weight update using the Delta Rule

Given

* initial weight (w) = 0.5
* Learning rate (η) = 0.1
* Input value (x) = 2
* Actual output (y) = 0.6
* Target output (t) = 1

The delta rule is a learning method used to update the weight of a neural network. It reduces the difference b/w the actual output and the target output

Formula:

$$\epsilon = t - y$$

$$\epsilon = 1 - 0.6$$

$$\boxed{\epsilon = 0.4}$$

$$\Delta w = \eta(t-y)x$$

Δw = change in weight

η = learning rate

$t - y$ = output error

x = input value

$\Delta w = 0.1\,(1-0.6)(2)$

$= 0.1 \times 0.4 \times 2$

$\boxed{\Delta w = 0.08}$

$w_{new} = w_{old} + \Delta w$

$w_{new} = 0.5 + 0.08$

$\boxed{w_{new} = 0.58}$

3. Confusion matrix performance Evaluation of a Disease prediction model

A confusion matrix is used to evaluate the performance of a classification model. It compares the actual results with the predicted results

Given

* True positive (TP) = 180
* True negative (TN) = 150
* False positive (FP) = 20
* False Negative (FN) = 50

$N = TP + TN + FP + FN$

$N = 180 + 150 + 20 + 50$

$\boxed{N = 400}$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$= \frac{180+150}{400} \times 100$$

$$= \frac{330}{400} \times 100$$

Accuracy = 82.5%

$$\text{Precision} = \frac{TP}{TP+FP} \times 100$$

$$= \frac{180}{180+20} \times 100$$

$$= \frac{180}{200} \times 100$$

$\boxed{\text{Precision} = 90\%}$

$$\text{Sensitivity} = \frac{TP}{TP+FN} \times 100$$

$$= \frac{180}{180+50} \times 100$$

$$= \frac{180}{230} \times 100$$

$\boxed{\text{Sensitivity} = 78.26\%}$

$$= \frac{150}{150+20} \times 100$$

$$= \frac{150}{170} \times 100$$

$\boxed{\text{Specificity} = 88.24\%}$

4. Genetic Algorithm for maximizing

A Genetic Algorithm (GA) is an optimization technique inspired by natural selection. It uses selection crossover, and mutation to find better

Given

Fitness function

$$f(x) = x^2$$

Initial population

$$P = \{2, 4, 6, 8\}$$

$f(x) = x^2$

$x = 2$

$f(2) = 2^2$

$= 4$

$x = 4$

$f(4) = 4^2$

$= 16$

$x = 6$

$f(6) = 6^2$

$= 36$

$x = 8$

$f(8) = 8^2$

$= 64$

* Individual $x = 8$ fitness $= 64$
* Individual $x = 6$ fitness $= 36$

$P_1 = 1000$

$P_2 = 0110$

$C_1 = 1000 = 8$

$C_2 = 0110 = 6$

The offspring remain unchanged in this example because the exchanged bits are identical

$1000 \rightarrow 1001$

$f(9) = 9^2$

$= 81$

5. overfitting in a Deep Neural Network

During the training of a deep neural network, training accuracy and validation accuracy are used to evaluate the learning performance of the model

Given

* Training accuracy increases from 60% to 95%.
* validation accuracy initially increases from 58% to 72%.
* Later, validation accuracy decreases from 72% to 68%.

1. Drop out

Dropout randomly disables a portion of neurons

during training.

2. Early stopping

* Early stopping monitors validation performance after each epoch. when validation performance stops improving traning stopped is retained

Significance of validation

1. It measures how well the model performs on data not used to update
2. It helps identify overfitting during traning
3. It helps selects the best epoch and model
4. It gives an indication of how well the model may generalize to unseen data.